

Ariadne

One question. Answers from many sources.

Federated knowledge for cybersecurity

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ReliaQuest Innovation Challenge
Challenge Three: Knowledge Graph Integration · October 19, 2026

What is a knowledge graph?

Relational Database

Structured records

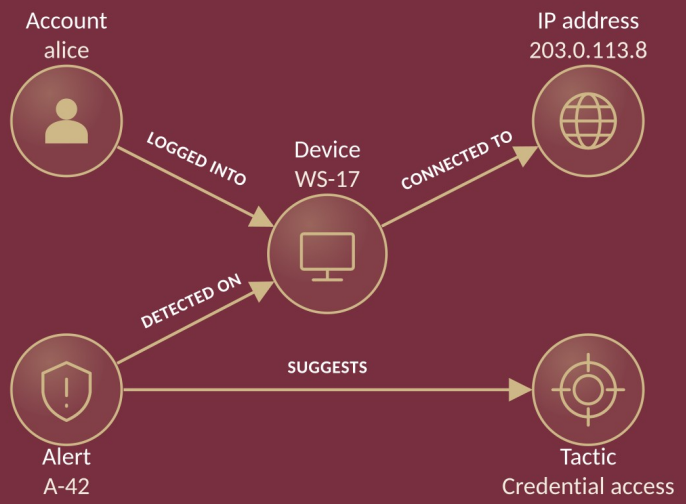
Authentication events

Time	Account	Device	Result
09:01	alice	WS-17	Failed
09:02	alice	WS-17	Failed
09:03	bob	WS-22	Failed
09:04	alice	WS-17	Success

How many failed logins per account?

Knowledge Graph

Heterogeneous entities



What activity connects this account to a threat?

1. **Entities become nodes**
   an account, a device, an alert, a tactic
2. **Connections are named**
   logged into · connected to · detected on · suggests
3. **Questions become paths**
   follow the links instead of joining tables

A table answers "how many". A graph answers "what connects".

Knowledge graphs speak different languages



Cypher

Attacks

```
MATCH (a:Attack)
WHERE a.technique = 'T1003'
RETURN a.name
```

中文



Gremlin

Mitigations

```
g.V().has('Mitigation',
  'technique', 'T1003')
 .values('name')
```

αβ



GQL

Logs

```
MATCH (e:Event)
FILTER e.technique = 'T1003'
RETURN e.timestamp
```

あい

Connect attacks, mitigations, and logs across graphs.

1 **MITRE ATT&CK**

adversary techniques and mitigations

Neo4j · speaks Cypher

2 **MITRE CAPEC**

attack patterns

Gremlin Server · speaks Gremlin

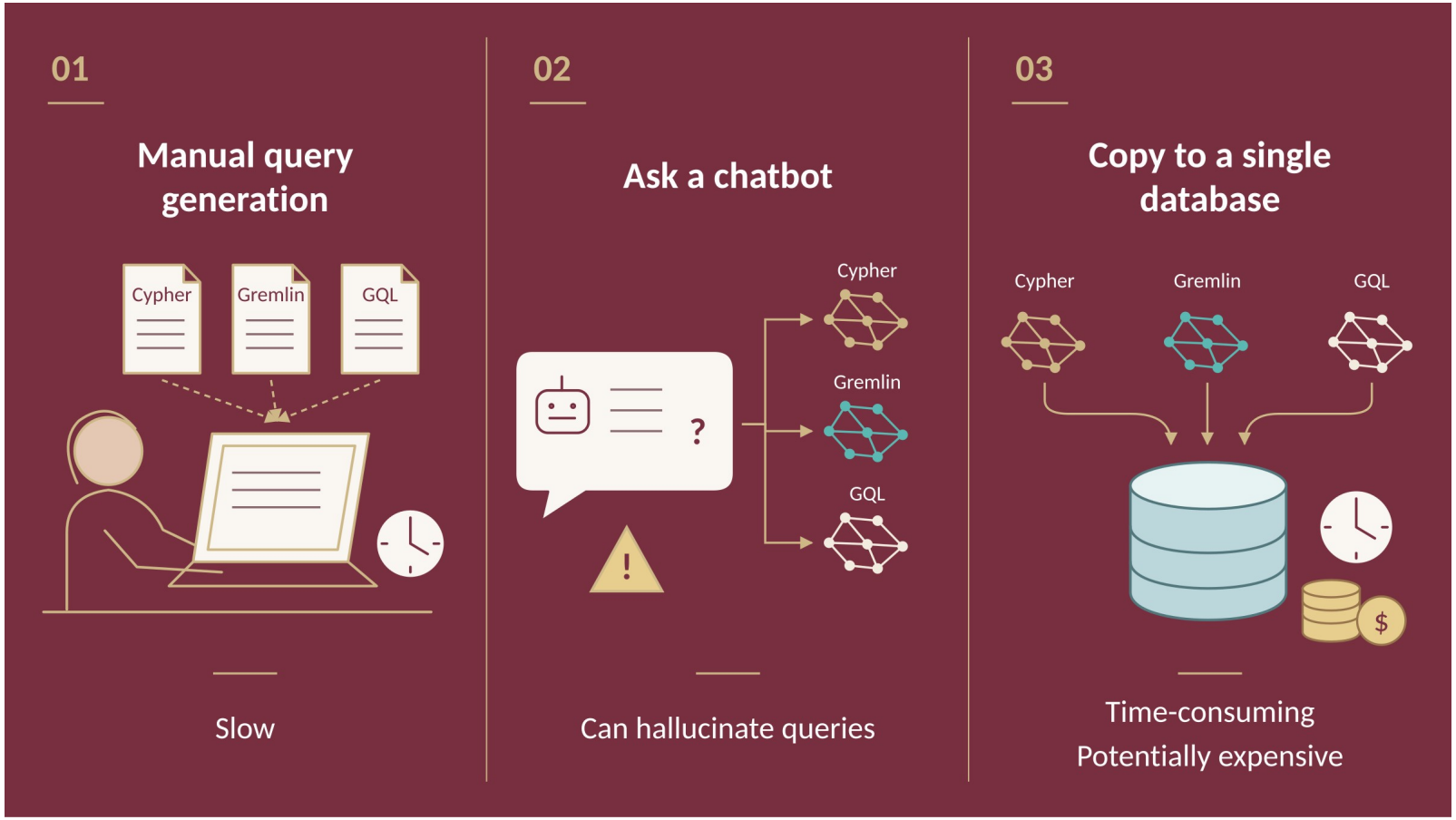
3 **Cloud graph databases**

Spanner Graph, Amazon Neptune

speak GQL and openCypher

Same question, three grammars.

How people query across graphs today

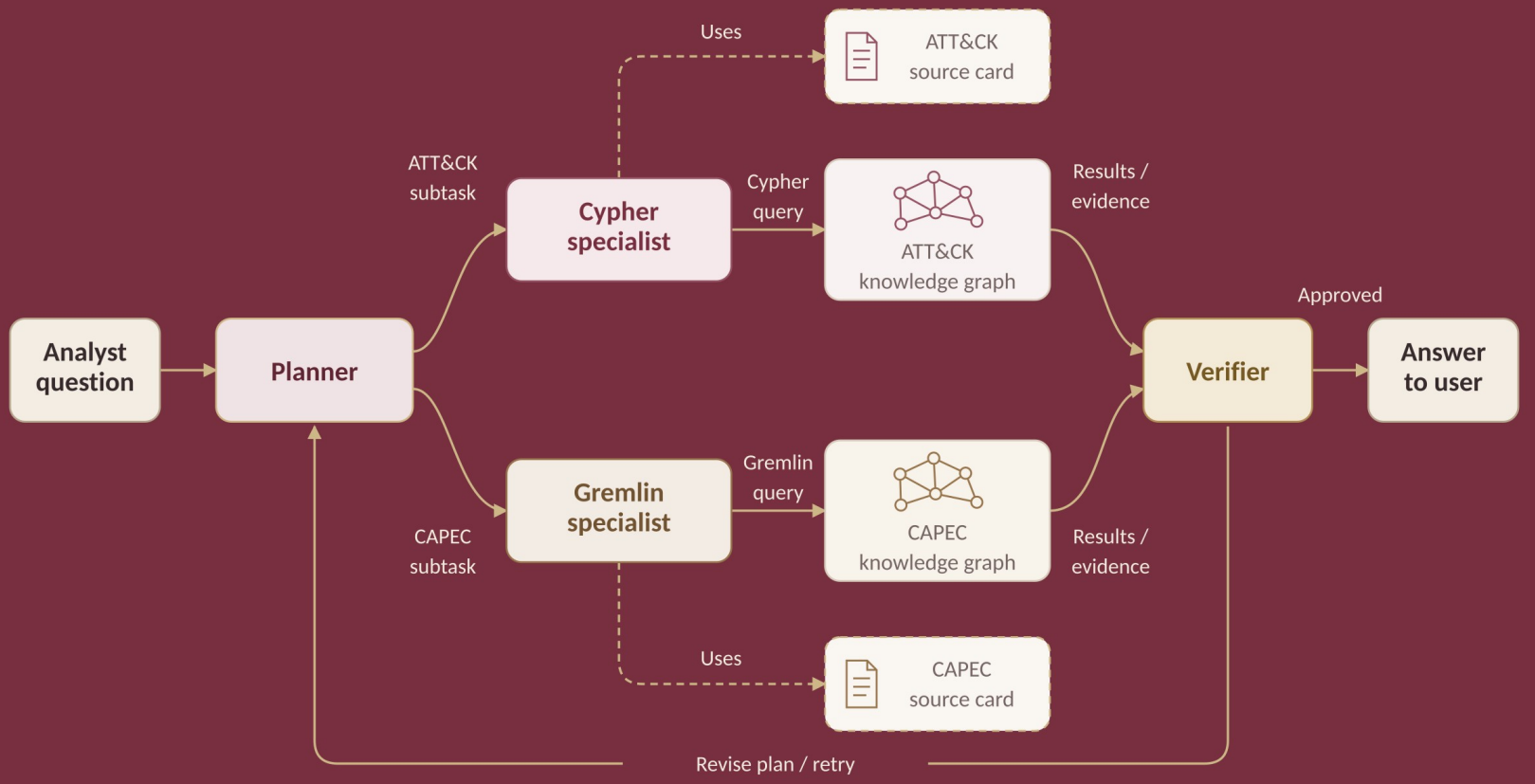


1. **Write it by hand**
   accurate, but slow and expert-only
2. **Ask a chatbot**
   fast, but the query is hidden and can be invented
3. **Copy into one database**
   one language, but expensive, stale, and it rewrites the sources

Hallucination: *an AI states as fact something that is not in the data.*

Each one trades away accuracy, speed, or visibility.

How Ariadne answers one question



Two rules

1 No model ever writes a query.

2 No claim without a record behind it.

One specialist per database. Each reads its source card and composes the query in that database's own language.

Evidence insufficient? The verifier sends the planner back, up to 5 rounds.

The source card: one sheet per database

`attack.json` the ATT&CK source card

- **Entities and properties**
  what the graph contains
- **Relationships**
  how entities connect
- **Join keys**
  how records link across databases
- **Query limits**
  read-only · depth · rows · timeout
- **Provenance**
  which snapshot, pinned by hash

39 KB of card describes a 19.9 MB graph.

What the specialist does with it

1. **Resolve** the question's entities against the card
2. **Compose** the query from five schema-derived primitives
3. **Gate** it: read-only, allowed labels, bounded depth and rows
4. **Execute** once, under a timeout
5. **Normalize** every record with its provenance

The verifier: every claim carries its record

- **Cited**
  every claim points at retrieved records
- **Resolvable**
  every identifier appears in those records
- **Grounded**
  what the claim says is in the record
- **Correlated**
  cross-database links are backed by the join index

What fails is cut and flagged. Deterministic code, not a model.

When the data has nothing

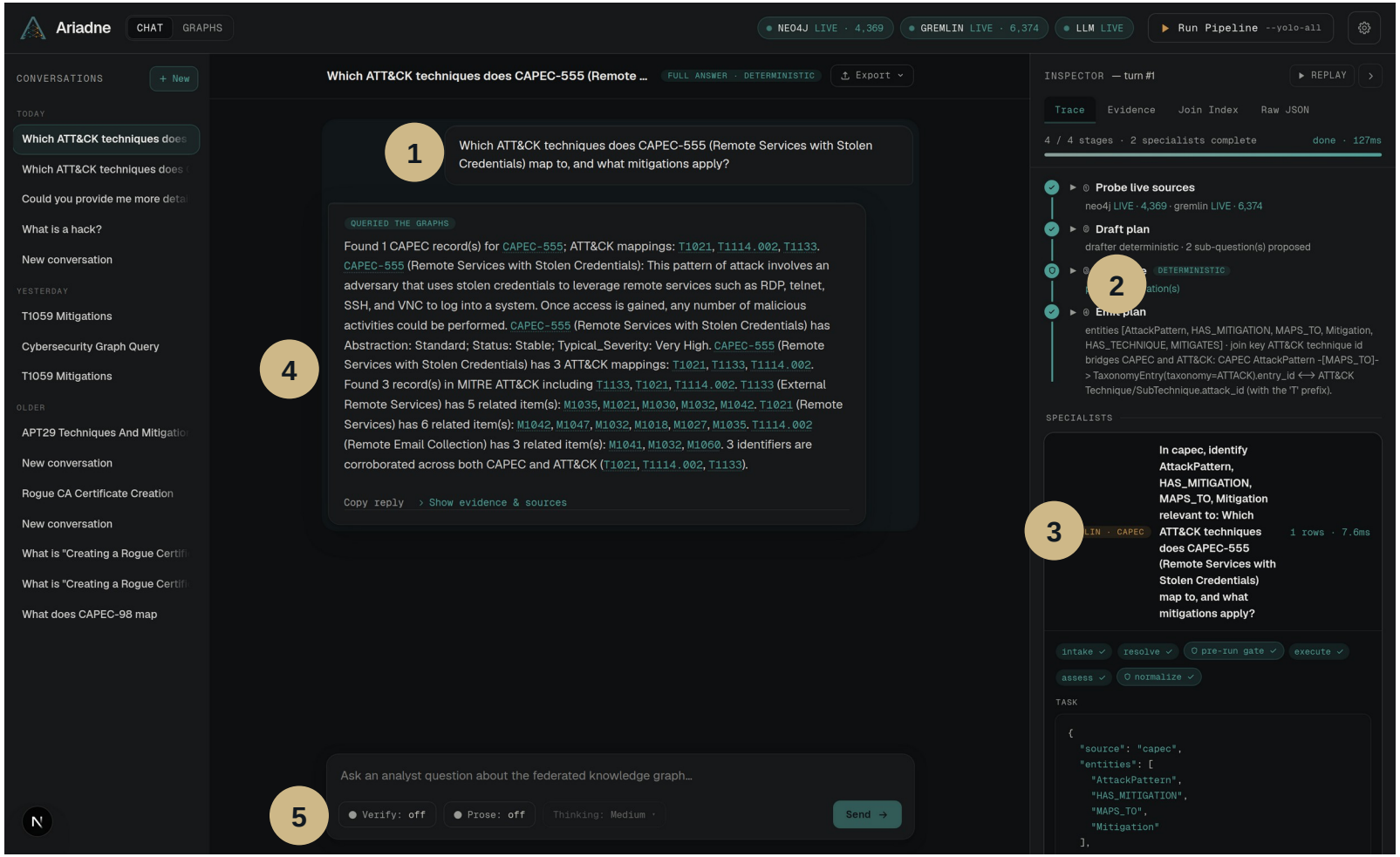
"The capec source doesn't establish this. The executed query ran cleanly and returned no matching records."

```
g.V().hasLabel("AttackPattern").has("id", within(ids))
 .out("MAPS_TO").hasLabel("TaxonomyEntry") …
```

0 claims made · 0 invented mappings · the query is the proof

An honest no, with the receipt.

The workbench: ask, watch it run, open the receipts



1. **The question**
   plain English

2. **The plan and its gate**
   which sources, which sub-questions

3. **One lane per database**
   the validated query, in its own language

4. **The answer**
   every identifier is a citation

5. **Three controls, off by default**
   Verify · Prose · Thinking: model help, per question

It scales across databases

3 · 2 · 1

databases · query languages · one process

Adding the third changed nothing about the questions.

0 vs 1,083

records rewritten: federating vs merging

Merging ATT&CK and CAPEC into one store rewrote 1,083 records. Federating rewrote none.

240 / 240

adversarial queries blocked before execution

After we found and closed 15 bypasses ourselves.

Adding a database is a source card, not a project.

Next: Amazon Neptune and Google Spanner Graph.

It already runs on ReliaQuest's own schema

What ReliaQuest sent

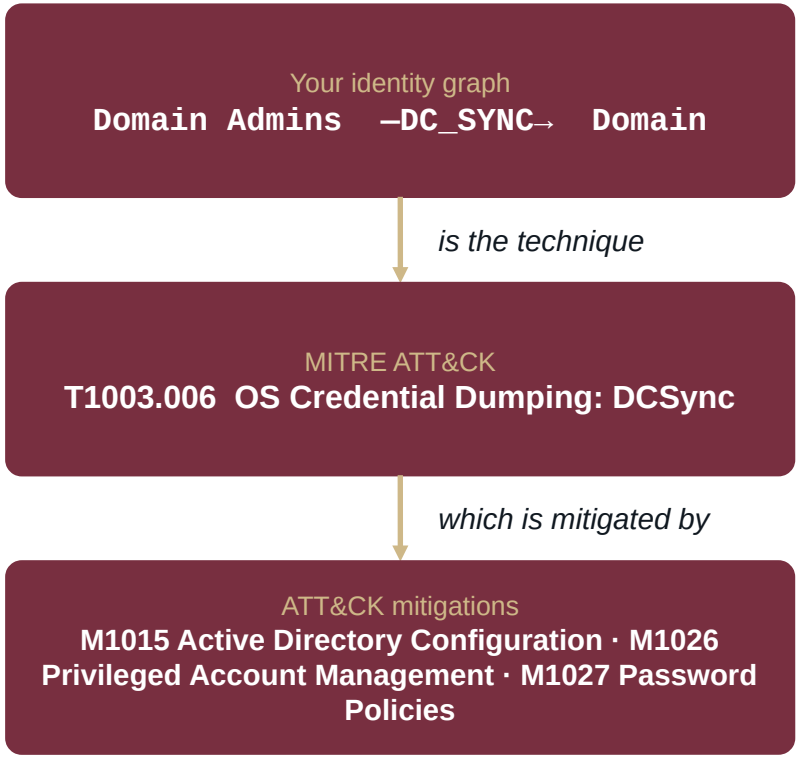
72
node types

21
relationship types

5
identity systems: AD · Entra · Okta · AWS · GCP

0
customer records needed

The bridge to ATT&CK



13 of 21 relationship types map to a technique; the other 8 are declared not to.

What happened

- The schema alone was enough. Records are synthetic; the schema is real.
- Ran live beside ATT&CK and CAPEC. Nothing else changed.
- A relationship in your graph resolves to a technique, then to its mitigations.
- Where no mapping exists, it says so.

"Which technique is what Domain Admins can do with DC_SYNC, and what mitigates it?"

Where it stops today

1	**Ask by identifier** "What mitigates T1003?"	**Works**
2	**Reword the question** same entity, new phrasing	**About half**
3	**Paraphrase the name** no identifier	**About a third**
4	**Describe it without naming it** no identifier, no name	**Not yet**

Also open

- Answer quality through the workbench's own planner is not yet judged.
- Identity records were synthetic; the schema was real.
- The planner, not the engine, sets the ceiling. It is the next target.

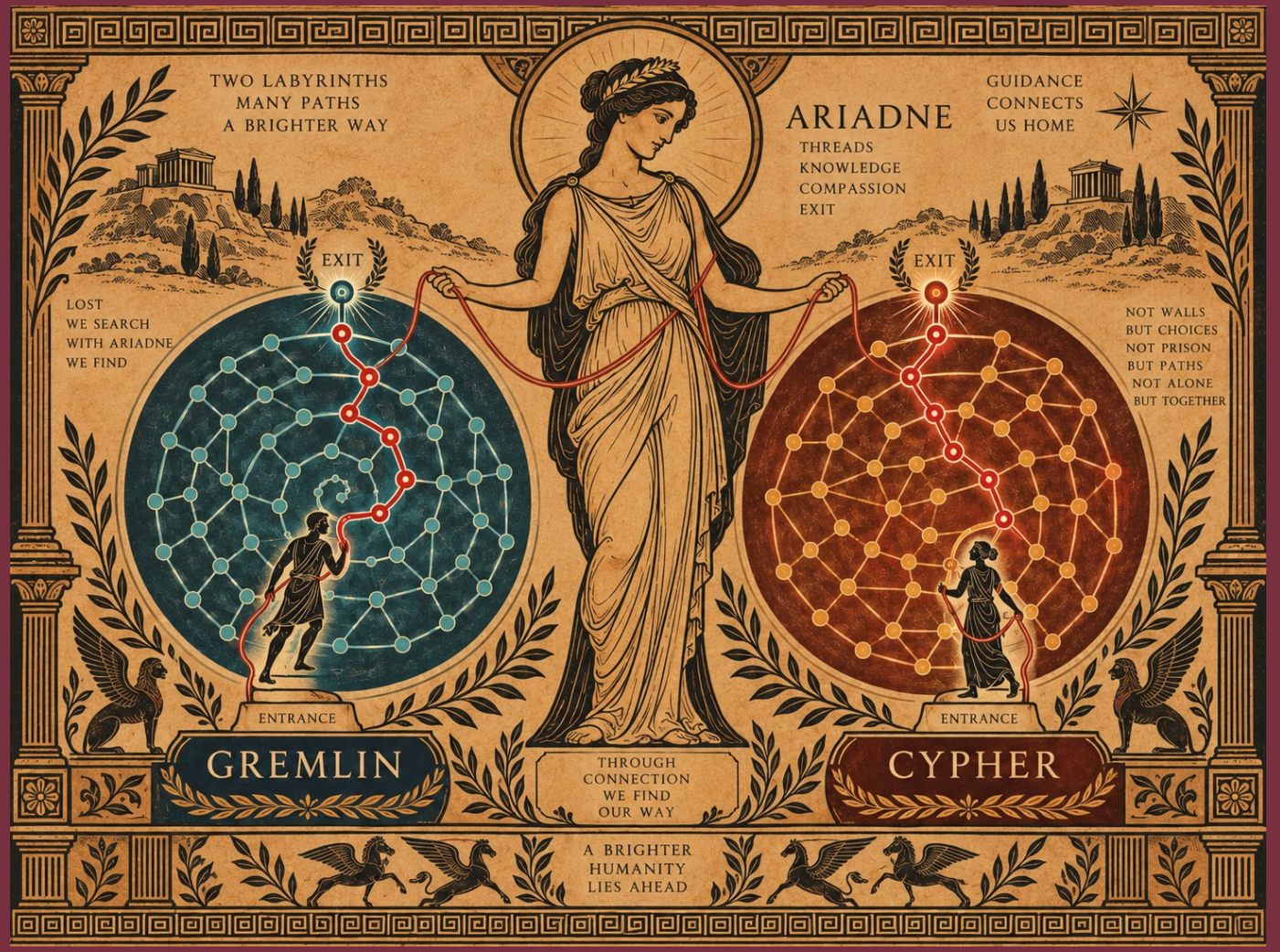
We know the next gap precisely, and it is measurable.

Thank you

Questions?

One question. Answers from many sources.

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github.com/brendondgr/Ariadne



Ariadne's thread guides travelers through the Gremlin and Cypher labyrinths.

Works cited

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Prior and related work

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